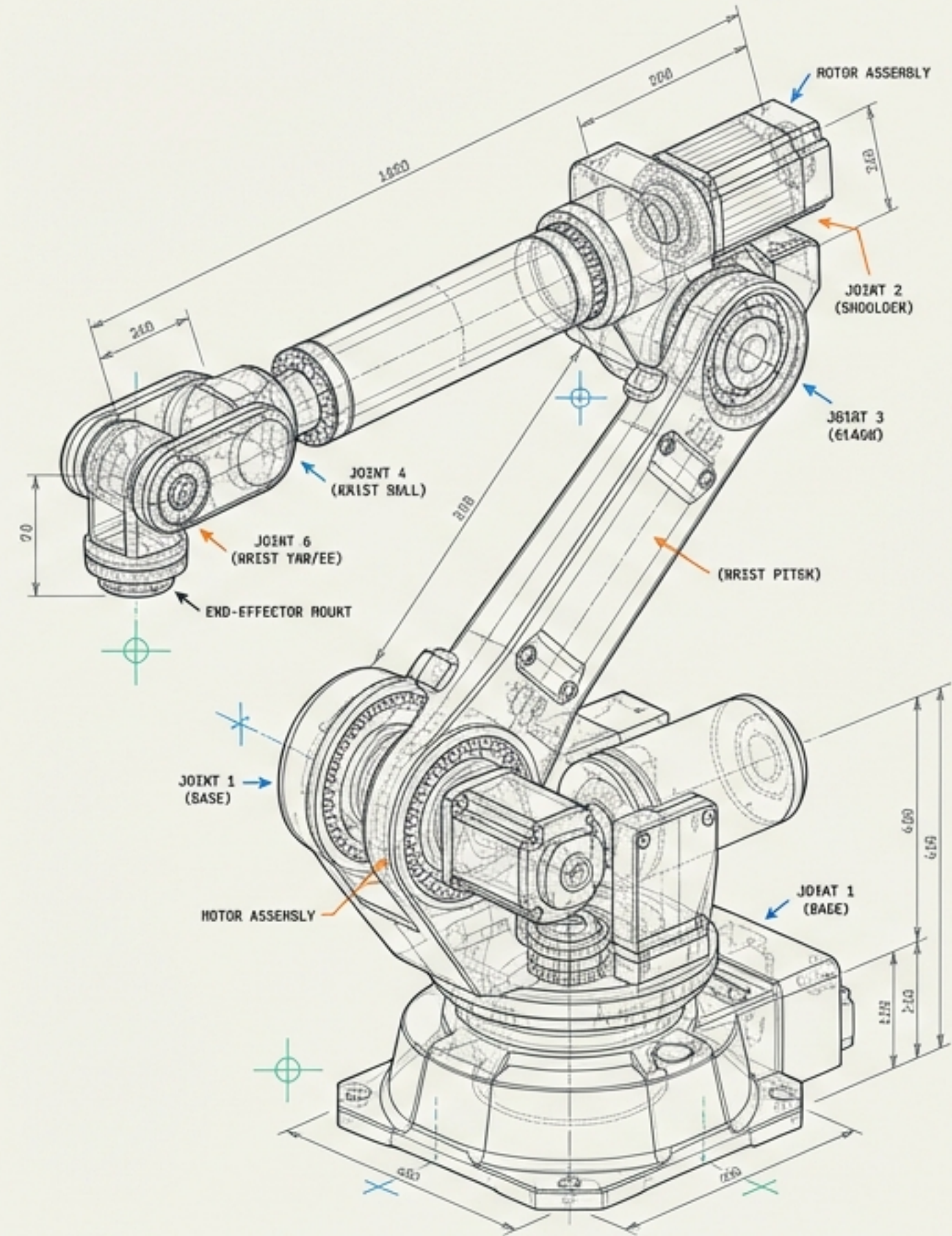


SO-101: The Optimization Journey

A Narrative Evaluation of Robotic Arm Learning, Failures, and Feature Stability.

DATE: 2026-01-31
STATUS: COMPLETED
SCOPE: 11 EXPERIMENTS (1 BASELINE + 10 VARIATIONS)



Executive Summary: The Efficiency Paradox

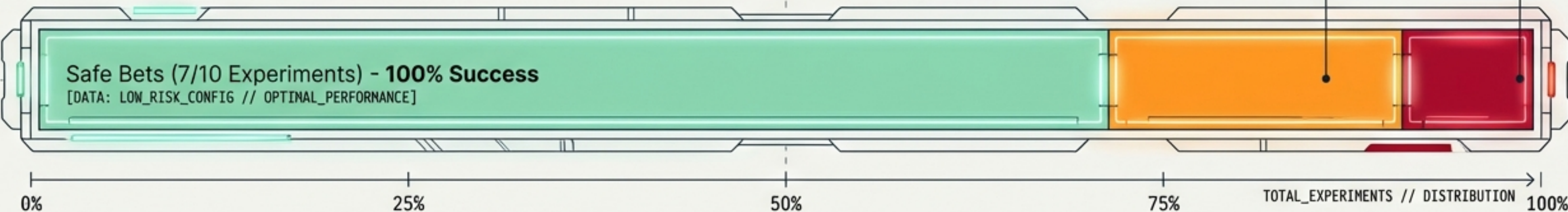
Bottom Line Up Front

“ The baseline configuration achieved a **100% success** rate. The investigation revealed a critical insight: **More isn't always better, but Less can be fatal.**

[REF: INVESTIGATION_LOG_001A // KEY_INSIGHT]

Success Distribution

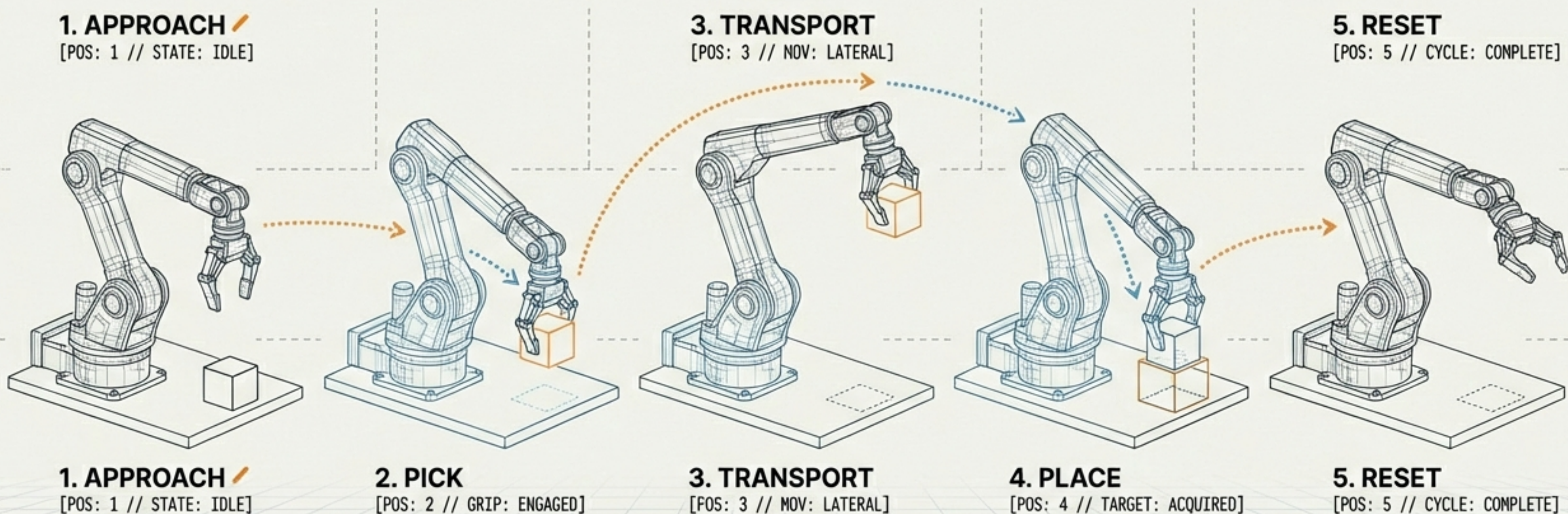
ANALYSIS: SUCCESS_RATE_DISTRIBUTION // Q1_RESULTS



TOTAL_EXPERIMENTS: 10 // SUCCESSFUL: 9 // FAILED: 1 // OVERALL_EFFICIENCY: 90% // REPORT_ID: FOR-SNI-MOD-001

The Mission & The Machine

The Task: The robot arm must learn a pick-and-place operation based on only 10 demonstration episodes. **The Research Question:** Can we improve upon perfection?



EXPERIMENT_ID: FOR-ROBOT-P&P // DEMONSTRATION_EPISODES: 10 // TASK: PICK-AND-PLACE // OBJECTIVE: EFFICIENCY_OPTIMIZATION

Decoding the Data: A Field Guide

Sub-headline technical forensics cumiplaing slide in Inter

Success Rate

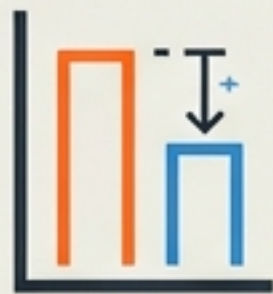
Like a batting average.



[DATA: PERFORMANCE_METRIC // KIT_RATE]

Cohen's d

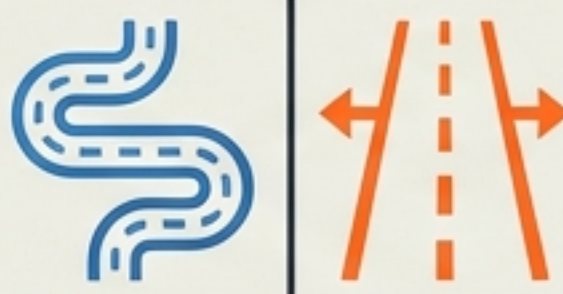
Signal vs Noise (Height comparison).



[DATA: EFFECT_SIZE // VARIANCE_RATIO]

Chunk Size

Turn-by-turn vs Highway planning.



[DATA: SEQUENCE_LENGTH // CONTEXT_WINDOW]

Temporal Ensemble

Video stabilization.



[DATA: TIME_SERIES // CONSISTENCY_CHECK]

VAE

Chef with multiple recipes.



[DATA: GENERATIVE_MODEL // LATENT_SPACE]

Dropout

Studying with missing pages.



[DATA: REGULARIZATION // FEATURE_MASK]

Learning Rate

Walking blindfolded.



[DATA: OPTINIZATION_STEP // GRADIENT_DESCENT]

KL Weight

Dress code strictness.



[DATA: LOSS_FUNCTION // REGULATION_TERN]

JetBrains Mono // TIME: 14:35:01 UTC // STATUS: CLASSIFIED_INTERNAL // CASE_FILE: 4N81-D

The Baseline: The Gold Standard

Metric Anchor Point

100%

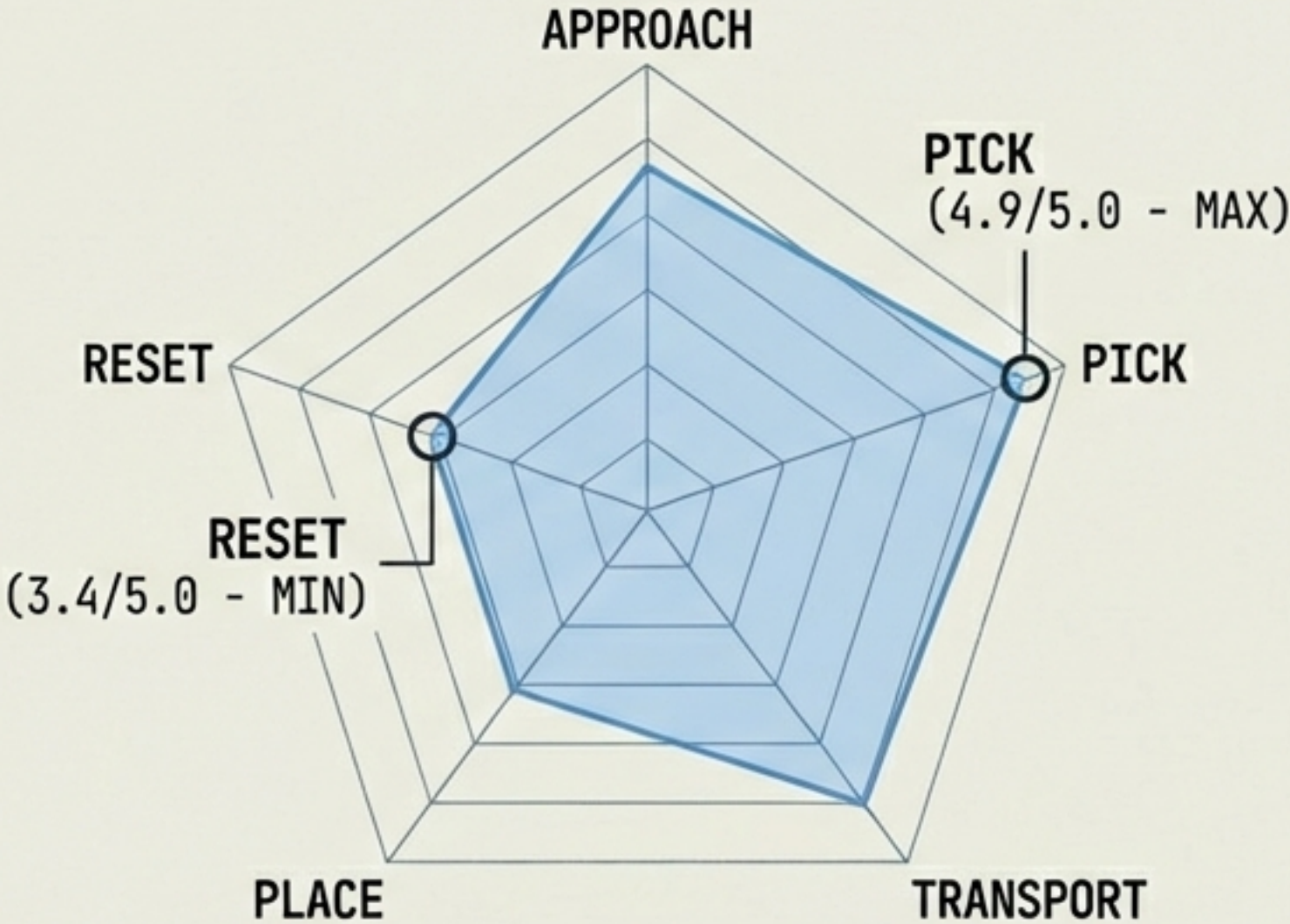
Success Rate (10/10 Trials)

Observer Note: "Well this is the baseline."

Observer Note: "Well this is the baseline."

This is the measuring stick. Every experiment asks: Is this better or worse than the baseline?

Phase Score Breakdown



Investigation: The Safe Bets (Top Performers)

Changes that maintained 100% success and improved qualitative feel.

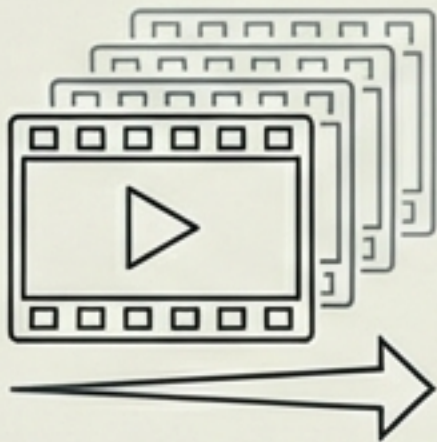
i3-temporal-ensemble

THE GOAT

SCORE: PERFECT

5.0

across all 5 phases.



[DATA: TEMPORAL_SMOOTHING // STABILITY_METRIC]

Insight:

Temporal smoothing created the most stable movement.



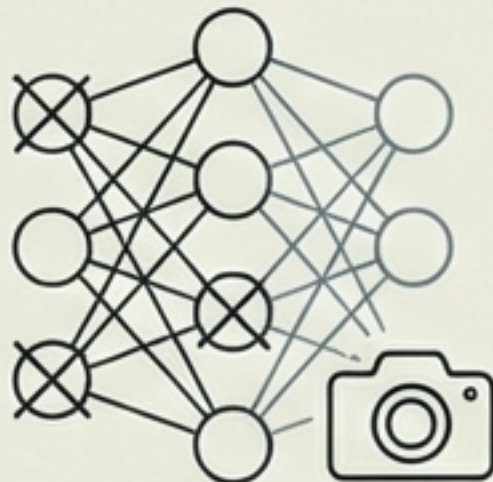
i2-dropout20-resnet50

ROBUST

SCORE: PERFECT

5.0

across all 5 phases.



[DATA: PERFORMANCE_SCORE // PHASE_CONSISTENCY]

Insight:

Successfully combined 'risky' parameters (Dropout + Vision).



Investigation: The Safe Bets (Stability Check)



7 out of 10 experiments matched the baseline.

[DATA: VARIATION_STABILITY // PHASE_ANALYSIS]



7/10 EXPERIMENTS //
STABILITY_VALIDATED



- ✓ **i1-chunk100-kl-low:** "Near perfect performance."
- ✓ **a3-chunk100:** "Created 'smoother motion' (Observer note)."
- ✓ **f1-lr-low:** "Proved safe convergence even with smaller learning steps."
- ✓ **h1-kl-low:** "Observer noted a 'more natural feel.'"
- ✓ **b2-no-vae:** "Removing VAE resulted in 'less jerky' movement."

[DATA: VARIATION_STABILITY // PHASE_ANALYSIS]

Key Insight: The model has headroom. It adapts well to parameter shifts without breaking.

[DATA: VARIATION_STABILITY // PHASE_ANALYSIS]

The Upgrade Trap: When Better is Worse

The Paradox

Case Study: d2-resnet50

Change: Upgraded vision from ResNet18 to ResNet50.

**Success
Dropped to 80%**

Diagnosis:

Better vision helped grasping but hurt placement. The bottleneck wasn't vision; it was fine motor control.

[DATA: UPGRADE_REGRESSION // VISION_BOTTLENECK]

The Visual Proof



[DATA: PHASE_SCORES_REGRESSION // PLACEMENT_FAILURE]

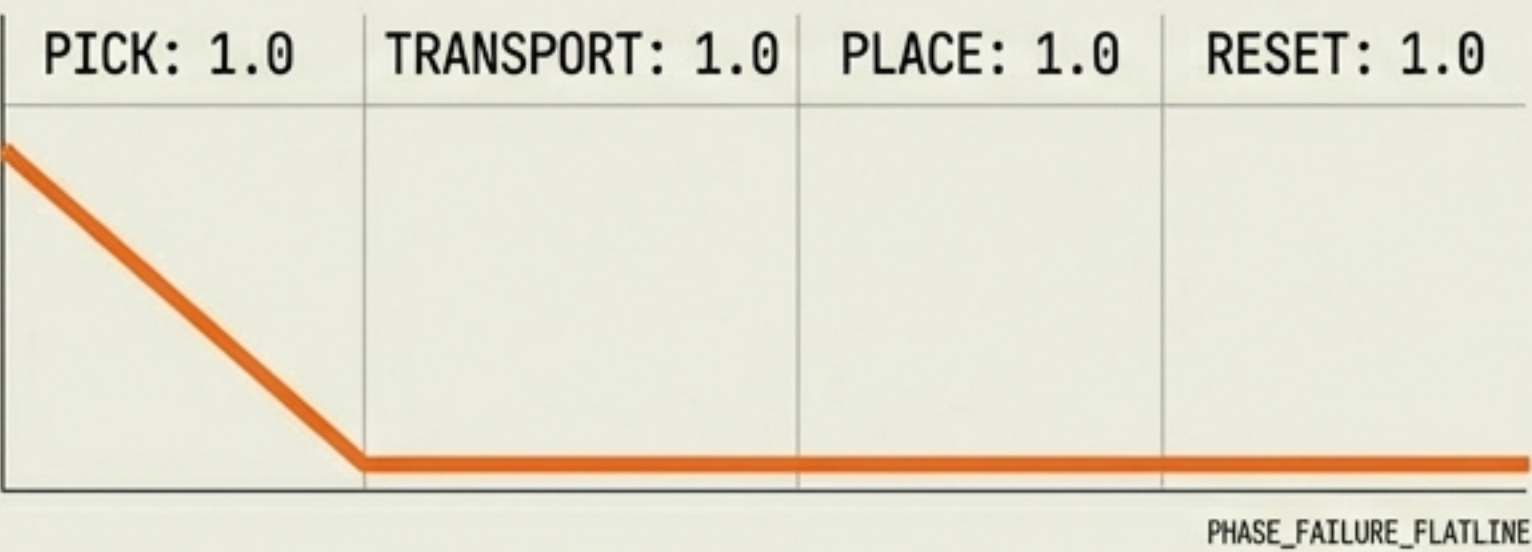
Bottom Note

Secondary Case: e3-dropout20 (Increased dropout) -> 90% Success.

Diagnosis: 'Feels too sensitive.'

[DATA: VARIATION_SENSITIVITY // DROPOUT_EFFECT]

The Capacity Floor: Catastrophic Failure



OBSERVER QUOTE:

“Total failure! Very fast movement and knocks off the object.”

ANALOGY:

“Trying to summarize a novel in one sentence.”

EXPERIMENT: c1-light // PARAMETER CHANGE:
Model Dimension reduced 512 → 256



LESSON:

“Parameter reduction has hard thresholds. Going below the floor causes collapse.”

[DATA: VARIATION_FAILURE // PHASE_ANALYST]

Anatomy of Failure: The 'Place' Problem

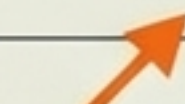
Forensic Phase Analysis in Inter

Approach	Pick	Transport	Place	Reset
Successful Experiments			Avg 4.3	
d2-resnet50 (Regression)			2.6	
Failed Experiments			Avg 1.0	

[DATA: PHASE_ANALYSIS // FAILURE_RAPPING]

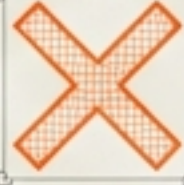
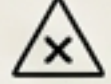
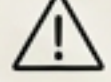
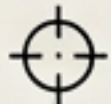


The Diagnostic Indicator.
The robot is great at Grabbing (Pick) but struggles with Letting Go (Place).



[DATA: PHASE_ANALYSIS // FAILURE_MAPPING]

Checking Intuition: Theory vs. Reality

Experiment	Prediction	Reality	
ResNet50	Better object recognition.	Regression (Wrong).	 
c1-light	Better generalization.	Catastrophic Failure (Wrong).	  
Temporal Ensemble	Smoother motion.	Smoother Motion (Correct).	 
Lower LR	Slower but safer.	Matched Baseline (Neutral).	 

Intuition must be calibrated by experimentation. ML logic does not always translate to robotic physics.

Signal vs. Noise: A Statistical Reality Check

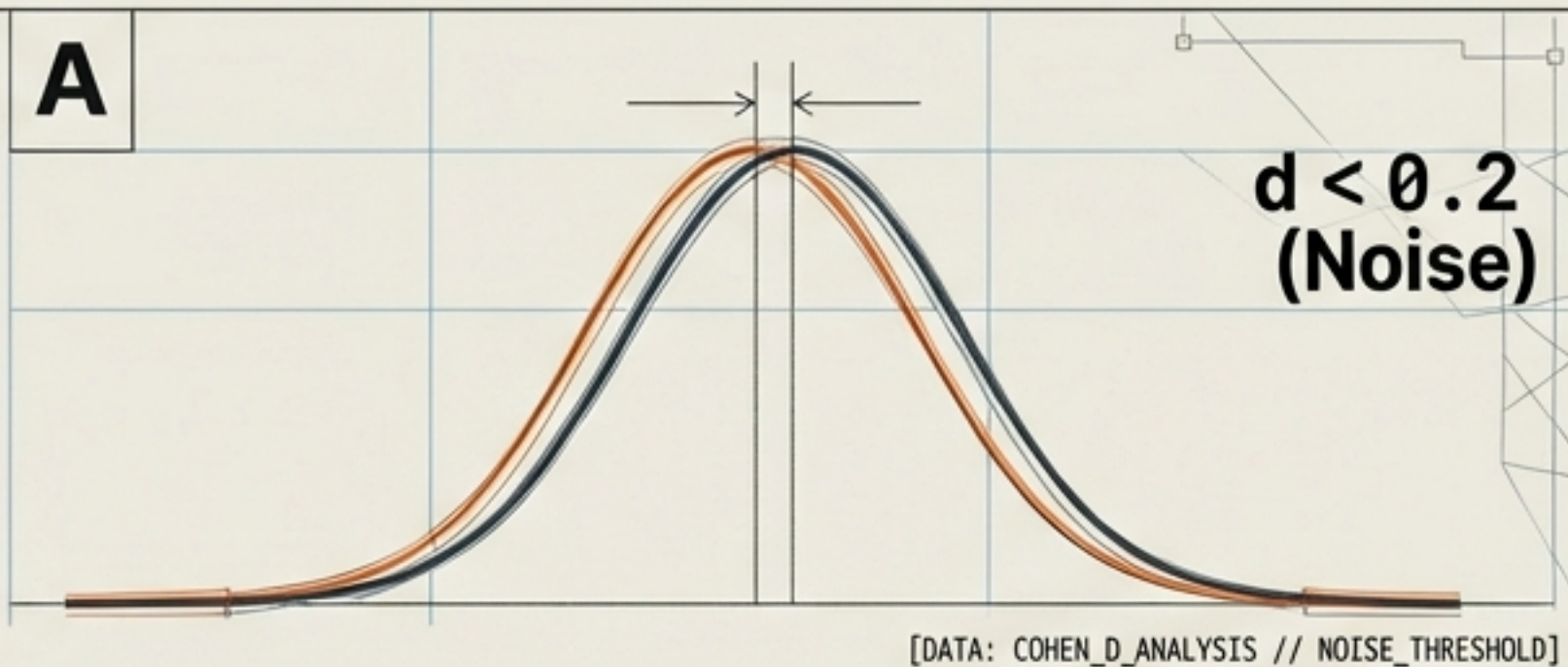
The Problem: With only 10 trials, is a 10% difference real or luck?

The Tool: Cohen's d (Effect Size).

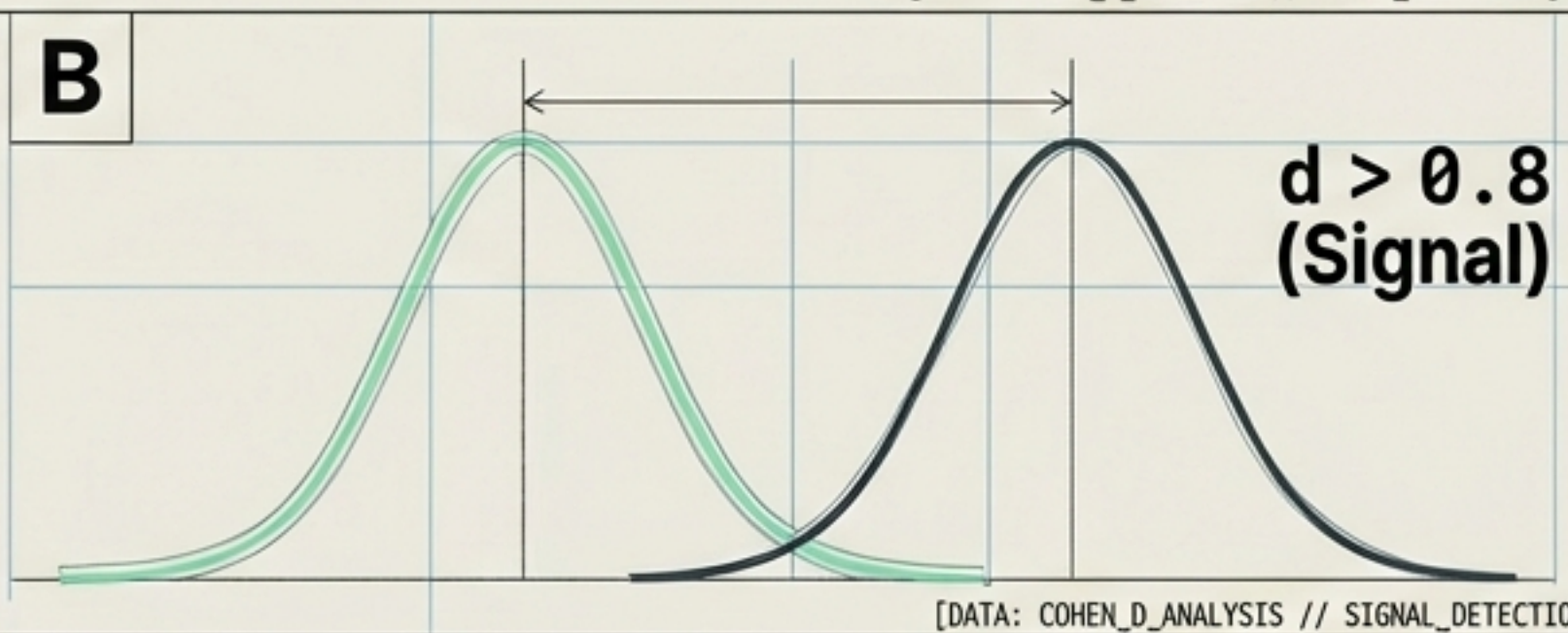
Trust Levels Legend

- $d < 0.2$: Negligible (Noise)** 
- $d = 0.5 - 0.8$: Medium (Investigate)** 
- $d > 0.8$: Large (Real)** 

A

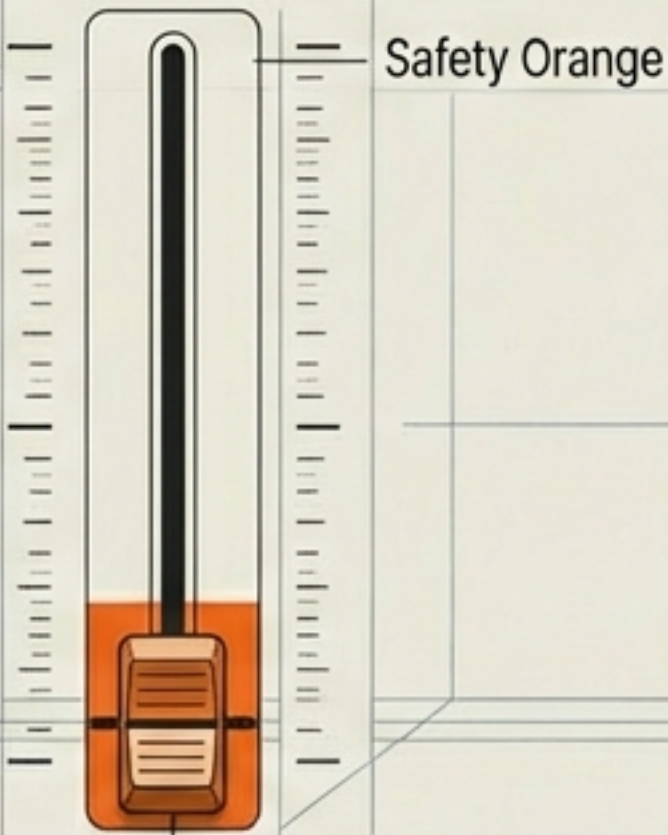


B



Control Levers: What Actually Matters

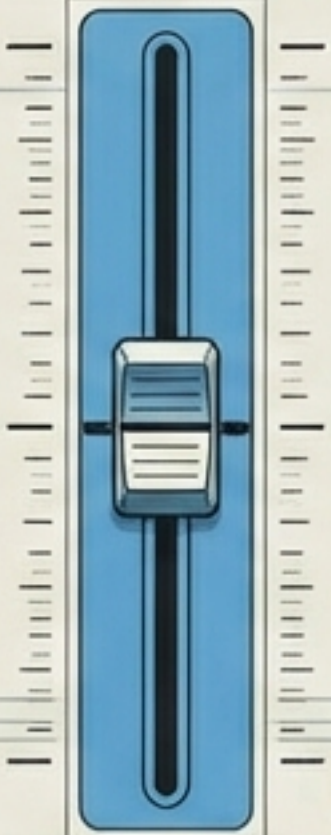
Model Capacity



[DATA: JETBRAINS MONO]

CRITICAL. Violation is fatal.
(dim_model ≥ 512)

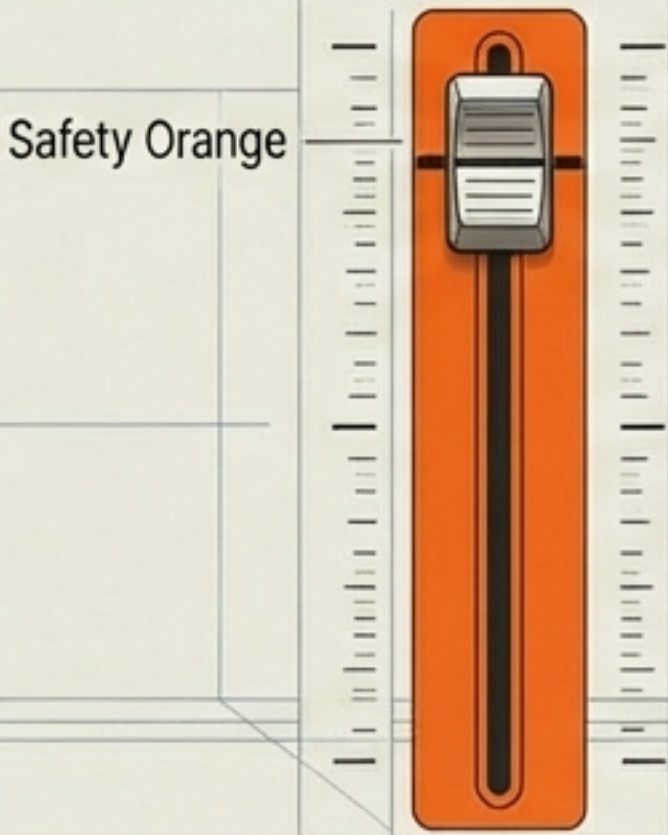
Tuning Parameters



[DATA: JETBRAINS MONO]

FLEXIBLE. Safe to tune: VAE,
Chunk Size, LR, KL Weight.

Complexity



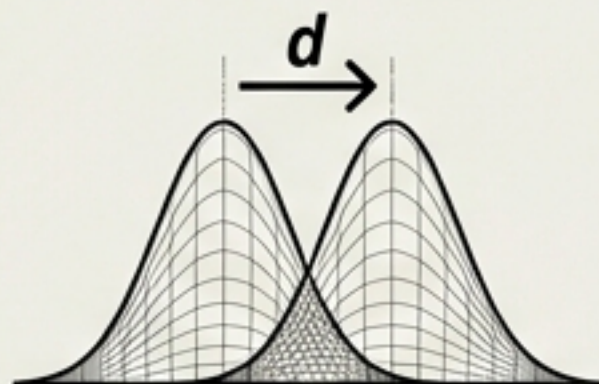
[DATA: JETBRAINS MONO]

RISKY. Proceed with caution:
Vision Backbone, Dropout.

Key Takeaways: Building ML Intuition

1 Effect Size Matters

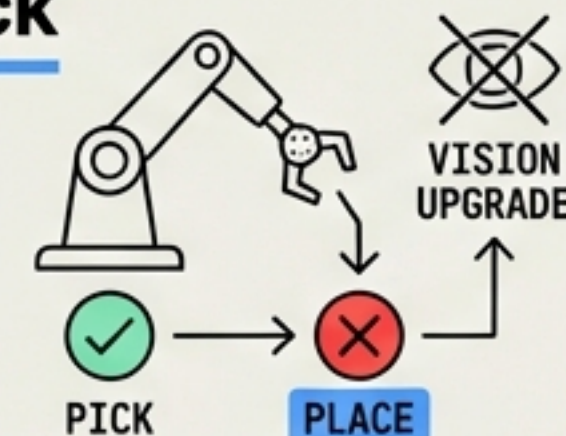
Trust patterns and effect sizes (Cohen's d), not just raw percentages.



[DATA: COHEN_D_ANALYSIS // PATTERN_VALIDATION]

2 Target the Bottleneck

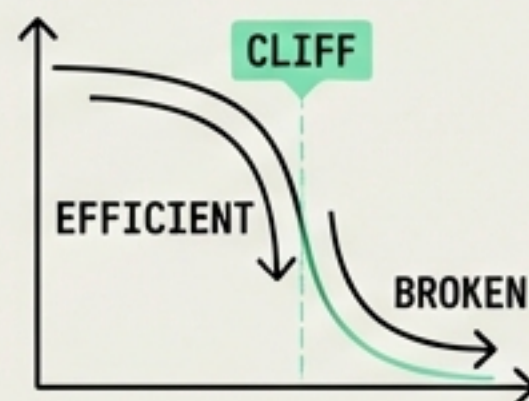
The system failed at Place, not Pick. Upgrading vision didn't fix the motor control issue.



[DATA: BOTTLENECK_ISOLATION // MOTOR_CONTROL_FAILURE]

3 Respect Thresholds

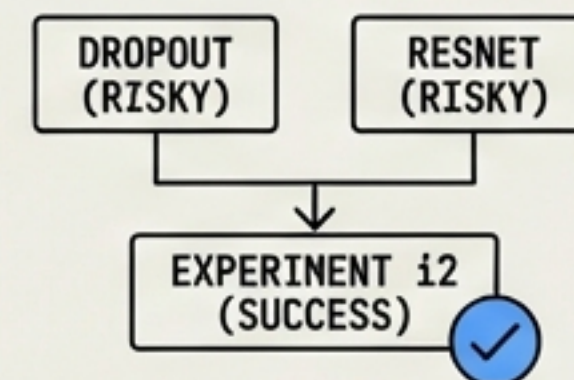
Parameter reduction has cliffs. The difference between "efficient" and "broken" is sharp.



[DATA: PARAMETER_CLIFFS // EFFICIENCY_BOUNDARY]

4 Combinations Can Surprise

Experiment i2 combined two "risky" parameters (Dropout + ResNet) yet succeeded.



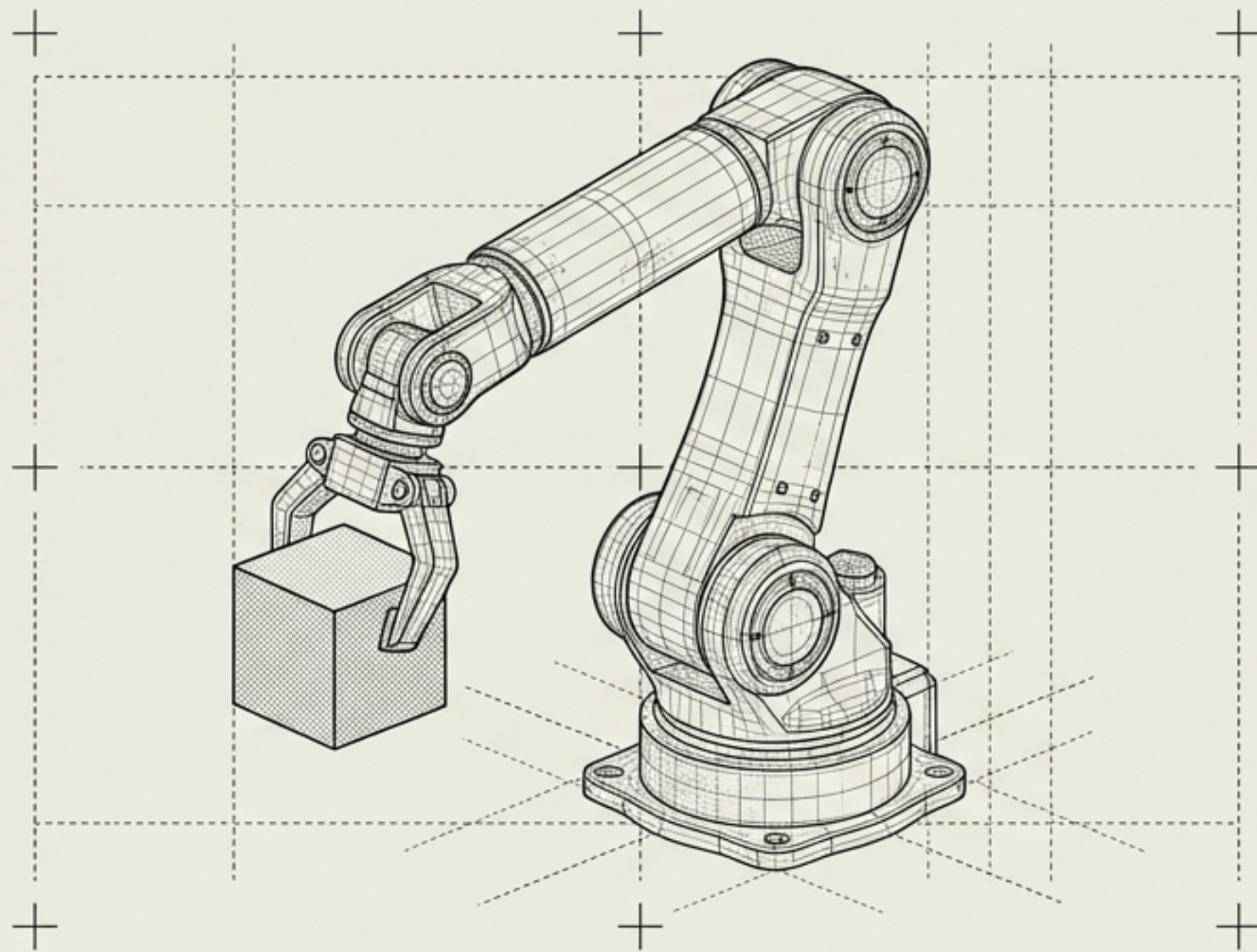
[DATA: UNEXPECTED_SUCCESS // COMBINATION_ANALYSIS]

The Verdict

“The baseline was robust not by accident, but because it met the capacity threshold without over-complicating the inputs.”

70% Safe Change Rate.

[DATA: BASELINE_SUCCESS // CAPACITY_THRESHOLD // INPUT_SIMPLICITY]



ACTIONABLE NEXT STEP

Proceed to test on a harder task variant.

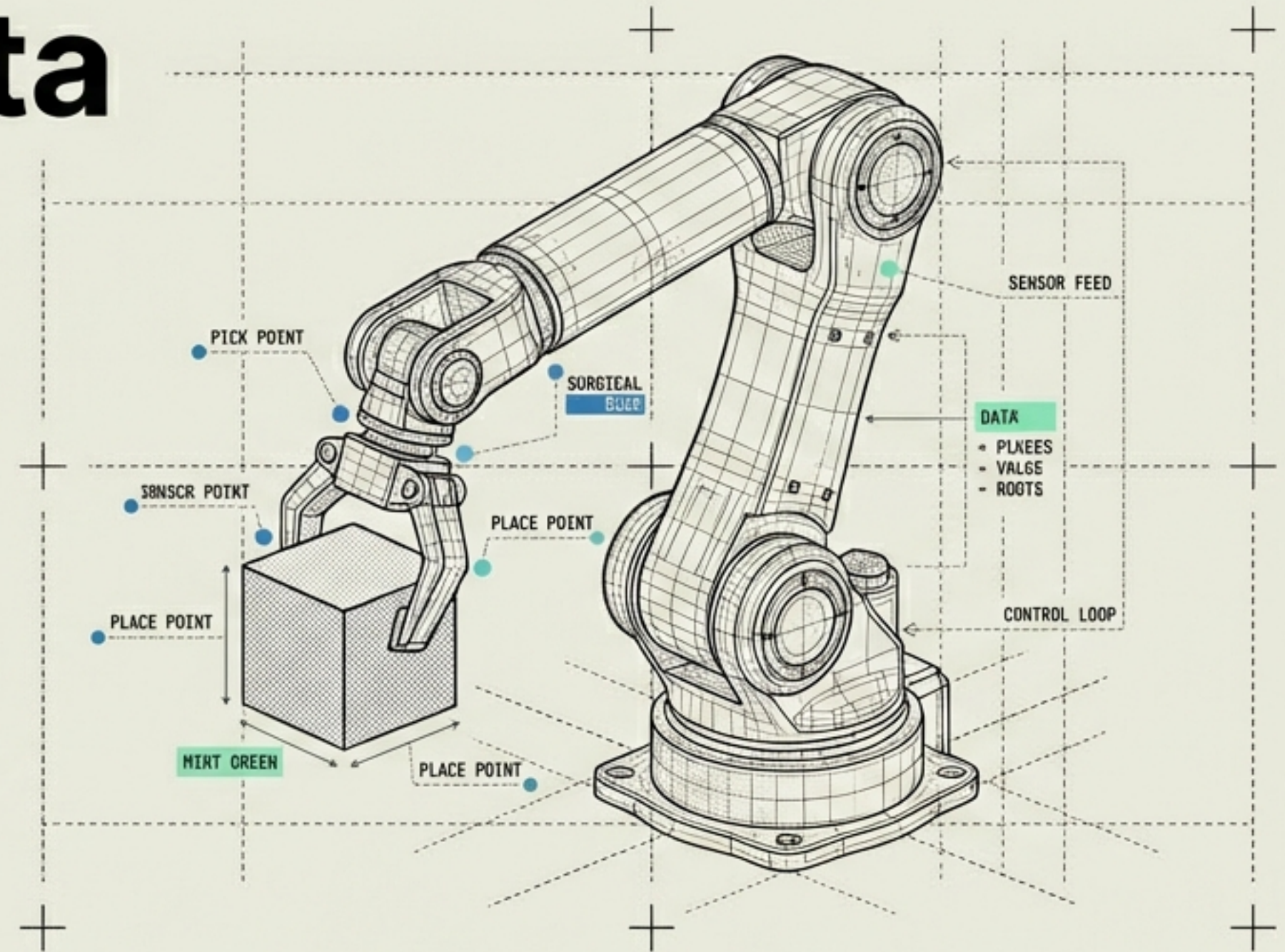
RATIONALE

Since 7 experiments hit 100%, the current task does not offer enough signal to differentiate 'Good' from 'Great'.

[DATA: EXPERIMENT_I2 // TASK_VARIANCE // SIGNAL_DIFFERENTIATION]

References & Data

- [DATA: SOURCE_REPORT // SO-101_EVALUATION]
Source Report: SO-101 Evaluation
Report (Generated 2026-01-31)
- [DATA: SYSTEM_SPEC // ROBOT_ARM_EVAL]
System: SO-101 Pick-and-Place Robot
Arm Evaluation System
- [DATA: METHODOLOGY // EXPERIMENT_DESIGN]
Methodology: 11 Experiments, 10 Trials
each, 10 Demonstration Episodes



The goal is building intuition, not just showing data.